

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 40 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Medium)

Here are the 10 exam questions as per your requirements:

Q2. [2 marks] (Easy)

What is the primary characteristic of a singly linked list?

Options:

A) Elements are stored in contiguous memory locations B) Each node points to its previous node C) Each node contains data and a pointer to the next node D) It allows direct access to any element by its index

Q3. [2 marks] (Easy)

Which data structure operates on the principle of Last-In, First-Out (LIFO)?

Options:

A) Queue B) Linked List C) Stack D) Tree

Q4. [2 marks] (Easy)

In a queue data structure, what operation is used to add an element?

Options:

A) Pop B) Enqueue C) Push D) Dequeue

Q8. [2 marks] (Medium)

What is the worst-case time complexity to search for a specific element by its value in an unsorted singly linked list?

Options:

A) $O(1)$ B) $O(\log N)$ C) $O(N)$ D) $O(N \log N)$

Section 2: Short Answer (6 Questions - 30 Marks)

Q5. [5 marks] (Medium)

Briefly explain two main advantages of using a linked list over a fixed-size array for storing a collection of elements.

Q6. [5 marks] (Medium)

Describe a real-world scenario where a Stack data structure would be an appropriate choice. Explain why it fits the stack's characteristics.

Q7. [5 marks] (Medium)

What is the primary difference in how elements are ordered and retrieved between a simple queue and a priority queue?

Q9. [5 marks] (Hard)

You have a singly linked list. Outline the conceptual steps (no pseudocode, but clear logical steps) to delete a node that contains a specific data value. Assume the list can be empty, the node to delete can be the head, or the value might not be found.

Q10. [5 marks] (Hard)

Explain how a stack can be used to convert an infix arithmetic expression to a postfix expression. Provide a small example, showing the stack's state and the resulting postfix string at key steps.

Q11. [5 marks] (Hard)

Contrast a standard linear queue (implemented with an array) with a circular queue. What specific issue common to linear array-based queues does the circular queue design effectively resolve?

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